

Regional Solar Power Potential Prediction Based on AI in Response to Climate Change

ChangYong Jung
Caramella Inc.
Research Institute
Gwangju, Republic of Korea
pm@caramella.kr

Dogyun Kim
Chonnam National University
Dept. Artificial Intelligence
Convergence
Gwangju, Republic of Korea
root@ql.gl

Seungbum Ha
Chonnam National University
Dept. of Computer Engineering
Gwangju, Republic of Korea
sbha@riia.or.kr

Sungjin Park
LoginSeogwang
Research Institute
Gwangju, Republic of Korea
psj6763@gmail.com

Jongman Kim
Jeonnam State University
Dept. of Electricity and
New&Renewable Energy
Damyang, Republic of Korea
jmk@dorip.ac.kr

Abstract

This study analyzes the impact of climate change on regional solar power potential and proposes a method for predicting it using an AI model. Climate change can alter key factors such as solar radiation, temperature, and cloud cover, which significantly affect solar power generation. Accurate predictions of these changes are crucial for formulating strategies to expand renewable energy. Using climate scenario data from the Korean Meteorological Administration, this study builds a deep learning model to analyze solar power potential in various region. Future research will extend testing to diverse geographical areas to enhance the model's generalizability and accuracy in reflecting varying climate conditions.

CCS Concepts

• **Computing Methodologies** → Neural networks; • **Applied computing** → Forecasting; Earth and atmospheric sciences; • **Mathematics of computing** → Time series analysis; • **Hardware** → Renewable energy.

Keywords

solar power forecasting, meteorological data, regional solar power, neural network

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1 Introduction

1.1 Background and Motivation

Existing power generation potential prediction models are mainly based on fixed data from the past and do not sufficiently reflect changes in regional power generation potential due to climate change. In a situation where solar irradiance and temperature changes change unpredictably due to climate change, the need for a dynamic and accurate prediction model that can reflect these changes is emerging. In particular, as climate change accelerates, the ability to predict future power generation potential is becoming more important for power supply stability and energy strategy establishment.

Solar power generation is inherently dependent on climatic factors such as solar irradiance, temperature, and cloud cover. These factors are highly variable and are expected to change significantly as the climate continues to warm. Therefore, the ability to accurately predict the potential of solar power generation in different regions has become increasingly critical for long-term energy planning. Traditional prediction models, which rely heavily on historical data, are ill-equipped to handle the complex and dynamic nature of climate change. These models often assume that past climate patterns will continue into the future, leading to inaccuracies in forecasting solar power potential.

As the impacts of climate change become more pronounced, such as increasing temperature variability, changing weather patterns, and more frequent extreme weather events, the limitations of traditional models are becoming evident. For instance, regions that once experienced consistent solar irradiance may now face more erratic patterns due to increased cloud cover or heatwaves, both of which can significantly reduce solar power output. As a result, there is a growing need for more advanced prediction tools that can account for the rapidly evolving climate conditions and provide more reliable data for energy planning.

1.2 Challenges in Solar Power Prediction

Accurately predicting solar power generation is a complex task that involves multiple environmental, geographical, and technological

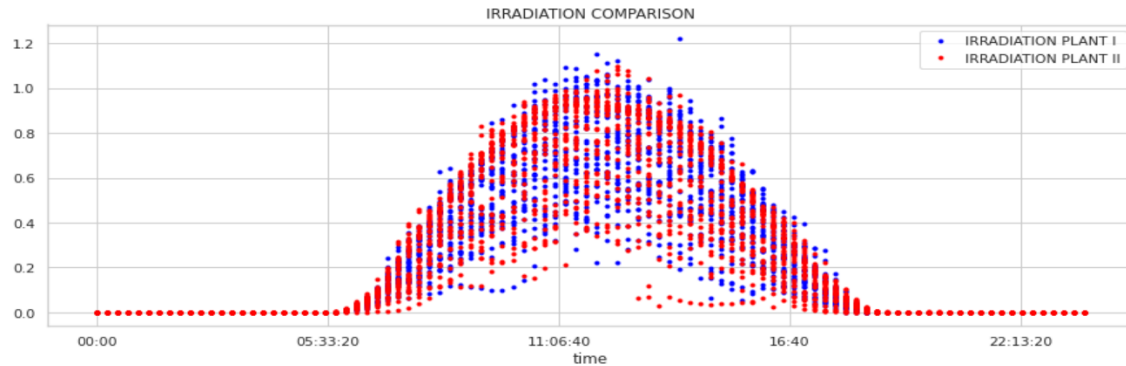


Figure 1: Irradiation Difference Between Nearest Plant

variables. One of the most significant challenges lies in the fact that solar power generation is highly sensitive to climatic conditions, which are increasingly being affected by climate change. As global temperatures rise, weather patterns shift, and extreme weather events become more frequent, traditional methods of predicting solar power potential are proving to be inadequate.

1.2.1 Variability in Solar Irradiation. Solar irradiance, the power per unit area received from the Sun, is the most critical factor influencing solar power generation. However, it is subject to significant variability due to cloud cover, atmospheric particles, and seasonal changes. Climate change exacerbates this variability by altering established weather patterns, leading to unpredictable fluctuations in solar irradiance.

For example, regions that previously enjoyed clear skies may now experience more frequent cloud cover, reducing the efficiency of solar power plants. Similarly, the occurrence of extreme weather events, such as heatwaves and storms, can further disrupt solar power generation; Traditional prediction models often fail to account for these rapid changes because they are based on historical weather data. These models typically assume that future weather patterns will follow the same trends observed in the past, which may no longer be true in a world experiencing the accelerated effects of climate change. As a result, predictions based on outdated climate data are likely to underestimate or overestimate the solar power potential in many regions.

1.2.2 Temperature Effects on Solar Panel Efficiency. Temperature is another critical factor influencing solar power generation. Although solar panels rely on sunlight, higher temperatures can decrease their efficiency. As the temperature rises, the electrical output of photovoltaic (PV) cells diminishes, reducing the overall power generation capacity. With climate change leading to more frequent and prolonged heatwaves, the efficiency of solar panels in certain regions could be significantly impacted. This presents a challenge for energy planners who rely on steady and predictable power output from solar installations.

Moreover, the relationship between temperature and solar panel efficiency is non-linear, making it difficult to model using traditional linear prediction methods. The need for more sophisticated

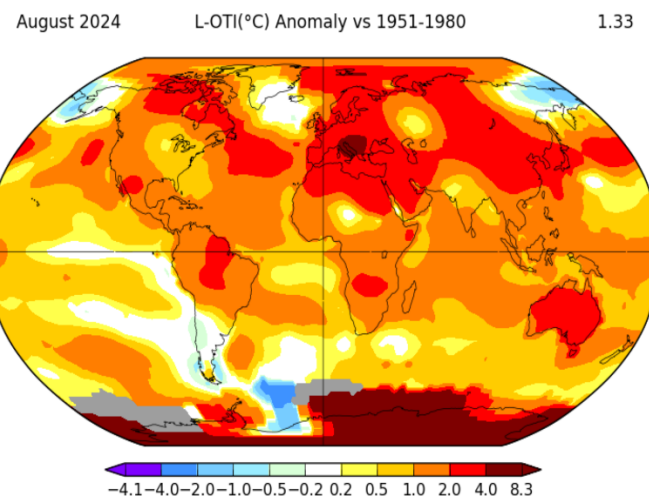


Figure 2: NASA Surface Temperature Analysis [2024.08 retrieves](<https://data.giss.nasa.gov/gistemp>)

models that can capture the complex interactions between temperature, irradiance, and solar panel efficiency is becoming increasingly evident.

1.2.3 Regional Differences in Climate Impact. Another challenge in solar power prediction is the regional variation in climate impacts. While some areas may experience an increase in solar irradiance due to climate change, others may see a decline. Factors such as geographical location, altitude, and proximity to large bodies of water all influence how a region will be affected by climate change. Therefore, a one-size-fits-all approach to solar power prediction is insufficient; Accurately forecasting solar power potential requires models that can account for the unique climate conditions of each region. This is particularly important for countries like South Korea, where climate conditions vary significantly across different regions. For instance, coastal regions may experience different weather patterns compared to inland areas, affecting the amount of solar energy that can be harnessed.

1.3 Purpose and Objectives of the Study

The primary purpose of this study is to develop an artificial intelligence (AI)-based model that can predict solar power generation potential across different regions by leveraging climate change scenario data. As the effects of climate change continue to alter key environmental variables such as solar irradiance and temperature, there is a growing need for more accurate and dynamic models capable of reflecting these changes. This study aims to fill the gap left by traditional prediction models that rely on static historical data and are unable to adapt to rapidly evolving climate conditions.

The model developed in this study will be trained using climate change scenario data provided by the Korea Meteorological Administration, which includes projections of temperature, solar irradiance, cloud cover, and other key variables over various time periods. Through machine learning techniques, the AI model will learn from these data to predict long-term solar power generation patterns with higher accuracy than traditional methods.

1.3.1 The specific objectives of the study. To analyze the impact of climate change on solar power generation potential; The study will explore how climate change affects critical factors such as solar irradiance, temperature, and cloud cover, which in turn influence solar power generation. By examining different climate change scenarios, the research will identify how these variables change over time and their potential effects on regional solar power potential; To develop a deep learning model for solar power potential prediction; The core objective of the study is to build an AI-based deep learning model capable of predicting solar power generation potential.

The model will be designed to incorporate climate change scenario data and adapt to evolving climatic conditions. This will allow for more accurate and reliable predictions of solar power potential in various regions; To evaluate the model's performance in predicting regional solar power potential; After developing the AI model, the study will evaluate its accuracy and reliability by comparing its predictions with actual solar power generation data. This evaluation will provide insight into the model's strengths and weaknesses and help refine the model for future applications.

2 Literature review

Research on predicting solar power generation has grown significantly, especially as climate change affects key environmental key factors like solar irradiance and temperature. This review focuses on the impact of climate change on renewable energy, the limitation of traditional prediction models, and the role of artificial intelligence (AI) in improving solar power forecasting.

2.1 Impact of Climate Change on Renewable Energy

Climate change directly affects solar power potential due to variations in solar irradiance and temperature. Hou et al. (2021) highlighted how some region in Europe could see changes in solar potential due to these variations. Similarly, Ben Othman et al. (2020) found that rising temperatures and increased weather variability in Tunisia could reduce photovoltaic (PV) efficiency. These studies emphasize the need for dynamic models that account for future climate conditions.

2.2 Limitations of Traditional Solar Power Prediction Models

Traditional models rely on historical data, which may not accurately reflect future climate conditions. Wild et al. (2021) argued that these models lack flexibility, making them ineffective in predicting solar potential under changing climate scenarios. Carvalho et al. (2023) called for advanced methodologies like machine learning to better capture the complex relationship between climate variables and solar power generation.

2.3 Utilization of Climate Scenario Data

Climate scenario data, such as those from IPCC models, offer valuable projections of future environmental conditions. Hou et al. (2021) and Ben Othman et al. (2020) showed that using these scenarios in prediction models improves accuracy and helps plan for regional differences in solar power potential. Integrating these data with AI models offers a powerful tool for accurate long-term forecasting.

3 data and methodology

3.1 Utilizing Climate Change Scenario Data for AI-Based Solar Power Prediction.

This study uses an AI model based on climate change scenario data [1] provided by the Korea Meteorological Administration to predict solar power generation potential by region. AI-based models can analyze large-scale data and learn long-term power generation pattern changes due to climate change to provide higher prediction accuracy. These predictions will provide important data for establishing more efficient renewable energy policies and selecting power plant locations by reflecting the changing solar power generation potential due to climate change.

Therefore, this study aims to contribute to future energy planning by predicting regional power generation potential according to climate change scenarios [1] by considering the impact of climate change on solar power generation..

3.2 Climate Change Scenario Data Provided by the Korea Meteorological Administration and Regional Solar Power Generation Data

The primary dataset for this study consists of climate change scenario data provided by the Korea Meteorological Administration (KMA). These data include a range of variables that influence solar power generation, such as temperature, solar irradiance, cloud cover, and precipitation [4]. These variables are projected under different climate scenarios, each representing possible future outcomes based on varying levels of greenhouse gas emissions.

3.3 Regional Solar Power Generation Data

The second key dataset consists of historical solar power generation data from various regions in South Korea. This dataset includes information on solar panel performance and generation capacity across different time intervals. The data covers multiple years and are structured to match the climate scenario data for corresponding regions.

Table 1: Climate Change Scenario Data (Extreme Climate Index)

Colum ID	Colum content	Datatype
TR25	Number of Hot Days	Number
HW33	Number of Heatwave Days	Number
SU25o	Number of Summer Days	Number
GSL	Plant Growth Period	Number
DTR	Diurnal Temperature Range	Number
TX90P	Warm Days	Number
WSDI	Duration of Warm Days	Number
WSDIx	Maximum Consecutive Warm Days	Number
TN90P	Warm Nights	Number
TXx	Maximum Daily High Temperature	Number
TNn	Minimum Daily Low Temperature	Number
FD0	Number of Frost Days	Number
ID0	Number of Ice Days	Number
CWm12	Number of Cold Days	Number
TX10P	Cold Days	Number
CSDI	Duration of Cold Nights	Number
CSDIx	Maximum Consecutive Cold Nights	Number
TN10p	Cold Nights	Number
TXn	Highest Maximum Daily Temperature	Number
TNx	Lowest Minimum Daily Temperature	Number

Table 2: Regional Solar Power Generation Data

Colum ID	Colum content	Datatype
field_seq	Site Number	String
sido_nick	Province Name	Number
date_no	Generation Date	Number
time_no	Generation Hour	Number
minutes_no	Generation Minute	Number
inverter_no	Inverter Number	Number
dc_v	DC Voltage	Number
dc_a	DC Current	Number
dc_w	DC Power	Number
ac1_v	AC1 Voltage	Number
ac2_v	AC2 Voltage	Number
ac3_v	AC3 Voltage	Number
ac1_a	AC1 Current	Number
ac2_a	AC2Current	Number
ac2_a	AC3 Current	Number
ac_w	Current AC Pow	Number
accum_energy	Power Generation by Time Interval	Number
to_energy	Daily Power Generation	Number
to_total_energy	Cumulative Daily Power Generation	Number
to_e_hour	Daily Operation Time	Number

Generation date and time: Daily and hourly solar power output. DC and AC power measurements: Direct and alternating current power outputs from solar panels. Cumulative generation: Total energy generated by the solar power system over specific time intervals. Operation time: Duration of solar panel operation, which helps in determining solar panel efficiency under different

conditions. By aligning these data with the climate change scenario data, the study creates a comprehensive dataset that captures both current solar power potential and predicted future climate conditions.



Figure 3: Result of regional solar power projection based on climate change scenario data.

3.4 Data Preprocessing

To ensure the datasets are suitable for model training, several preprocessing steps are undertaken:

Data Cleaning: Missing or inconsistent data are handled through imputation techniques or by removing invalid entries. For example, gaps in solar power generation records due to equipment malfunction are either filled using average values from similar time periods or excluded from the dataset if significant.

Normalization: Climate and solar power data are normalized to ensure all variables are on a comparable scale, which is necessary for improving the performance of machine learning algorithms. For instance, temperature data might be scaled between 0 and 1 using min-max normalization.

Time Series Alignment: Since the study focuses on predicting future solar power generation based on projected climate changes, it is essential to align the historical solar power data with the future climate scenario data. This is done by using common time intervals and synchronizing regional data points between both datasets.

Feature Engineering: Additional features are created to capture relevant patterns in the data. For example, the study includes derived variables like solar irradiance deviation from the monthly average and cumulative temperature changes over time.

3.5 AI Model Development

The core of this study is the development of an AI-based deep learning model designed to predict solar power potential using the processed data. A deep neural network (DNN) [5] architecture is chosen due to its ability to model complex, non-linear relationships between climate variables and solar power generation.

The model architecture consists of several layers:

Input Layer: This layer receives climate change scenario data, including variables like solar irradiance, temperature, cloud cover, time periods.

Hidden Layers: Multiple hidden layers with fully connected neurons process the input data. Each hidden layer applies activation functions (e.g., ReLU) to capture non-linear relationships between the input features and solar power output. These layers also help

the model learn interactions between multiple climate variables that affect solar power generation.

Output Layer: The output layer predicts solar power potential, either as a continuous value representing daily power generation or as a classification that indicates whether a region's solar power potential will increase or decrease in the future.

3.6 AI Model Training

The model is trained using historical solar power generation data and corresponding climate data, allowing it to learn patterns in solar power output based on environmental conditions.

The training process involves:

$$Loss = \frac{1}{m} \sum_{i=1}^m (P_{solar_i} - \hat{P}_{solar_i})^2 \quad (1)$$

Loss Function: A mean squared error (MSE) as in equation (1)' loss function is used to minimize the difference between the predicted and actual solar power generation values during training.

Optimizer: The Adam optimizer is employed for efficient gradient descent, which updates the model's parameters to minimize the loss.

Regularization: Dropout layers and L2 regularization are used to prevent overfitting, ensuring that the model generalizes well to new data.

3.7 AI Model Evaluation

The models' performance is evaluated using several key metrics:

Mean Absolute Error (MAE): This Measures the average absolute difference between predicted and actual solar power values.

Root Mean Squared Error (RMSE): A more sensitive error metric that penalizes larger errors more than smaller ones.

R² Score: This evaluates how well the model's predictions fit the actual data, providing an overall measure of model accuracy.

3.8 Simulation and Prediction

Once trained, the AI model is used to simulate future solar power potential in different region of South Korea under varying climate

change scenarios. These simulations predict how solar power generation will evolve over the next few decades, allowing for the analysis of:

Regional Disparities: How different regions may see increases or decreases in solar power potential based on projected climate changes. **Temporal Trends:** Predicted trends in solar power generation over time, highlighting periods of higher or lower generation potential. The simulation results provide valuable insights for policymakers, enabling them to make informed decisions on solar power plant locations and energy policy adjustments.

4 Conclusion

This study demonstrates the critical importance of dynamic, AI-based models for predicting solar power generation potential in the context of climate change. Traditional models relying on historical data fail to capture the rapidly evolving climatic variables such as solar irradiance and temperature [2]. By leveraging climate change scenario data from the Korea Meteorological Administration, the proposed AI model provides improved accuracy in predicting regional solar power potential [3]. These insights can guide the development of renewable energy policies, optimize the location of solar power plants, and contribute to energy strategy formulation, enhancing stability in power supply amid changing climate conditions.

However, the current study has limitations, as the model was tested and validated in only one region. To improve its generalizability and accuracy, future research will expand testing to diverse geographical areas, providing a comprehensive evaluation of the

model's effectiveness across varying climate conditions. Ongoing refinements will focus on adapting the model's parameters to specific regional climate patterns, thereby increasing its practical applications in global renewable energy planning.

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